

was at least 2 mm of margin between the plaque and tumor edges, and V_{100} values in the post-plans exceeded 98%. For one patient, the margins were not as generous and the V_{100} dropped from 100% to 58%.

Conclusions: This is the first study verifying the geometry and dosimetry of EBT in patients using MRI. Furthermore, by understanding the differences in estimated and administered implant geometries, planning guidelines can be made more robust to address potential shifts and tilts. Work is underway to implement heterogeneity-based dose calculations.

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Uveal Melanoma Treated With Iodine-125 Episcleral Plaque According to the COMS Trial Is Associated With Worsened Visual Acuity Change and Increased Risk of Chronic Toxicity Compared to Lower Prescription Doses

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Purpose/Objective(s): In the treatment of uveal melanomas, the optimal prescribed dose to maximize disease control, but minimize radiation-related complications is unknown. Historically our institution has treated uveal melanomas to doses less than 85 Gy to the tumor apex even if the apex was less than 5 mm in height. Here, we investigate how chronic toxicity and visual outcomes are affected by the radiation dose at the tumor apex.

Materials/Methods: A retrospective review was performed to evaluate patients treated for uveal melanoma with Iodine-125 plaques. Primary end points included visual acuity, radiation-related complications, and eye preservation. Univariate and multivariate analyses were performed to determine association between radiation dose and the end point variables.

Results: One hundred ninety patients with sufficient data to evaluate the end points were included. The median tumor height was 3.9 mm (interquartile range 3-5.4 mm) and the median maximal tumor diameter was 11.0 mm (interquartile range 9.5-13 mm). The median dose delivered to the tumor apex was 81.0 Gy (interquartile range 68.7-89.0 Gy). The median duration of follow-up was 4.1 years (interquartile range 2.4-5.9 years). Increasing apex dose and dose to 5mm depth were correlated with greater visual acuity loss ($p = 0.02$, $p = 0.0006$), worse final visual acuity ($p = 0.02$, $p < 0.0001$) and increased radiation complications ($p < 0.0001$, $p = 0.0009$). The 5 year actuarial risk of any radiation complication was 73%. The most common radiation complications were retinopathy and cataract formation. Enucleation rates were associated with higher dose to 5 mm ($p = 0.0001$).

Conclusions: We have identified a relationship between higher dose to tumor apex or 5mm and increased risk of visual acuity loss and radiation complications including enucleation, retinopathy, and cataracts. We have previously reported no effect on tumor control with doses as low as 69 Gy prescribed to tumor apex. Consideration should be given to prescribing 70 Gy to tumor apex, especially in the setting of significant contralateral ocular morbidity.

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Hypofractionated Stereotactic Radiation Therapy Using Helical Tomotherapy for Single or Multiple Brain Metastases

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Purpose/Objective(s): According to the American Society for Radiation Oncology's evidence-based guidelines on the management of newly diagnosed brain metastases, patients with multiple brain metastases have

several treatment options, including stereotactic radiosurgery (SRS) alone. However, hypofractionated stereotactic radiation therapy (SRT) seems to become more appropriate than single-fraction SRS with the increase in tumor size. The purpose of this study was to evaluate the clinical outcomes of a 4-fraction SRT study using helical tomotherapy for brain metastases.

Materials/Methods: Between August 2009 and June 2013, 54 patients with a total of 128 brain metastases underwent SRT using helical tomotherapy. The eligibility criteria were as follows: (1) brain metastases diagnosed with contrast-enhanced T1-weighted magnetic resonance imaging; (2) maximum tumor diameter ≤ 3 cm; (3) tumor number ≤ 10 ; (4) no cerebrospinal fluid dissemination; (5) imaging findings and clinical presentation consistent with brain metastases; (6) no concurrent chemotherapy; and (7) no previous whole-brain radiation therapy, SRS, or SRT to the target volume to be treated. The primary endpoint of the study was local control (LC) at 6, 12, and 18 months. Secondary endpoints were overall survival (OS) and freedom from distant brain failure (FDBF) at 6, 12, and 18 months, and the incidence of acute and late toxicities. A total dose of 28 or 28.8 Gy at 80% isodose was administered for all tumors in 4 fractions. The mean gross tumor volume (GTV) was 1.9 cc.

Results: Local control (LC) rates at 6, 12, and 18 months were 96%, 91%, and 88%, respectively. The 12-month LC rates for tumors with GTV ≤ 0.25 cc, > 0.25 and ≤ 1 cc, and > 1 cc were 98%, 82%, and 93%, respectively; the rates were 92% for tumors > 3 cc and 100% for > 10 cc. OS rates at 6, 12, and 18 months were 61%, 52%, and 38%, respectively. FDBF rates at 6, 12, and 18 months were 61%, 41%, and 27%, respectively. The 6-month rates for FDBF were 57%, 71%, and 55% for patients with 1, 2, and ≥ 3 brain metastases, respectively. All of these differences were not significant. There were no acute adverse events of grade 3 or higher. Only one patient had grade 2 nausea during treatment and one had grade 2 motor neuropathy. The late adverse events were grade 2 neurologic dysfunction observed in 2 patients (4%) and there was no radiation necrosis.

Conclusions: The 4-fraction SRT protocol provided excellent tumor control with minimal toxicity. Distant brain failure was not so frequent, even in patients with multiple tumors. The results of the current study warrant a prospective randomized study comparing single-fraction SRS with SRT in this patient population.

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Noninvasive Stereotactic Radiosurgery for Brain Metastases of Lung Cancer

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Purpose/Objective(s): To investigate the clinical outcomes of non-invasive stereotactic radiosurgery for brain metastases of lung cancer in our institution.

Materials/Methods: Sixty-seven patients with 109 brain metastases of lung cancer treated with non-invasive stereotactic radiosurgery between 1998 and 2011 were retrospectively analyzed. The median age was 63 years (range, 29-82). Sixty-four patients and 3 patients were previously diagnosed with non-small cell lung cancer and small cell lung cancer, respectively. Performance status by ECOG were 0-1 in 47 patients and 2-3 in 20 patients. Median Brinkmann index (BI) was 450 (range, 0-2580). The median size of metastatic lesions was 11 mm (range, 3- 55 mm) and the numbers of brain metastases were 1 in 37 patients, 2 in 19 patients, 3 in 7 patients, and ≥ 4 in 4 patients. Twenty-eight patients had uncontrolled primary disease and 34 patients had extracranial metastases. Seventy-six lesions were treated with a single fraction with the median dose of 24 Gy and 30 lesions were treated with 3 fractions with median doses of 30 Gy. The median follow-up time was 9.4 months (range, 0.4- 125). The in-field recurrence (IFR), intracranial recurrence (ICR) and overall survival (OS) rates were calculated using Kaplan-Meier method. Univariate (log-rank